

1. Explain the 'Right to Professional Conscience', 'Right to Conscientious Refusal', and 'Right to Professional Recognition'.

Right to Professional Conscience

- Engineers have the right to make decisions based on their moral and ethical values while performing their duties.
- If an engineer believes a certain action is unethical or harmful, they have the right to oppose it.
- Their decision cannot be challenged if they are acting within ethical guidelines.

Example:

An engineer working on a bridge project realizes that using low-quality materials may cause accidents in the future. Even though the management pressures them to continue, they refuse to approve the materials, ensuring public safety.

Right to Conscientious Refusal

- This is the right to refuse to participate in work that goes against ethical or moral principles.
- Engineers are not forced to do something they consider unethical.
- It applies when a large group agrees something is unethical, or even when there is a debate about it.

Example:

A software developer is asked to create an app that collects users' personal data without their knowledge. They refuse because it violates privacy rights.

Right to Professional Recognition

- Engineers should get proper credit for their work, such as patents, promotions, or appreciation.
- Recognition motivates engineers and encourages innovation.
- This can be internal (within the company) or external (industry awards, patents, etc.).

Example:

A mechanical engineer invents a new fuel-efficient engine. They receive a patent for their work, ensuring their contribution is recognized.

2. What do you mean by 'Intellectual Property Rights' (IPR), and why should they be protected? Briefly explain patents, trademarks, and copyright.

Intellectual Property Rights (IPR)

- IPR protects the creations of inventors, artists, and businesses from being copied without permission.
- It ensures that creators can benefit from their hard work.
- IPR promotes innovation and encourages businesses to invest in research.

Why protect IPR?

- Prevents theft of ideas and inventions.
- Encourages businesses to innovate.
- Helps companies make a profit from their unique products.

Patents

- A patent is a legal protection given to inventors to prevent others from making or selling their invention for a fixed period (usually **20 years**).
- It applies to **new inventions** that are useful and original.

Example:

The technology behind fingerprint unlocking in mobile phones is patented. Other companies cannot use it without permission.

Trademarks

- A trademark protects logos, brand names, and symbols that help identify a business.
- It ensures customers recognize the company behind a product.
- Trademarks can last **indefinitely** if renewed.

Example:

The **Nike swoosh** and **McDonald's golden arches** are trademarks, meaning no other company can use them.

Copyright

- Copyright protects **creative works** like books, music, films, and computer software.
- It lasts for the **lifetime of the creator plus 70 years**.

Example:

A musician records a song, and copyright ensures others cannot copy or sell it without permission.

3. What do you understand by 'Corporate Social Responsibility' (CSR)?

Support your answer with an example.

Corporate Social Responsibility (CSR)

- CSR means companies should not just focus on making money but also **help society** and **protect the environment**.
- Businesses should act ethically and support social causes.

Benefits of CSR:

- Improves the company's reputation.
- Helps in environmental protection.
- Builds customer trust and loyalty.

Example:

Tata Group spends a portion of its profits on education and healthcare programs, helping underprivileged communities.

4. Discuss how safety and risk should be an integral part of any process or design.

Safety and Risk in Engineering

- Engineers must always consider safety while designing machines, buildings, or software.
- If safety is ignored, it can lead to **accidents, financial loss, or harm to human life**.
- Risk assessment helps identify **potential problems** and **prevents disasters**.

Steps for Risk Assessment:

1. **Identify possible risks** – What could go wrong? (e.g., machine failure)
2. **Analyze the impact** – How dangerous is the risk? (e.g., system crash)
3. **Implement safety measures** – Find ways to reduce risk (e.g., add warning signals).
4. **Continuous testing and improvement** – Check safety regularly.

Example:

Car manufacturers **crash test** new models to ensure they are safe before selling them to customers.

5. What is a code of ethics? Elucidate the IEEE Code of Ethics.

What is a Code of Ethics?

- A **code of ethics** is a set of rules that professionals must follow to ensure honesty and fairness.
- It helps maintain integrity and ensures that engineers do their job responsibly.

IEEE Code of Ethics

The **Institute of Electrical and Electronics Engineers (IEEE)** has 10 key principles:

1. **Public Safety First** – Engineers should ensure their work does not harm people.
2. **Avoid Conflicts of Interest** – They should not take decisions that benefit them personally.
3. **Be Honest** – Engineers must not lie about test results or product safety.
4. **Reject Bribery** – Engineers must refuse money or gifts that can lead to corruption.
5. **Promote Ethical Use of Technology** – They should educate people about safe technology use.
6. **Stay Updated** – Engineers should keep learning to improve their skills.
7. **Accept Criticism** – Engineers should admit mistakes and correct them.
8. **Treat Everyone Fairly** – No discrimination based on race, gender, or religion.
9. **Respect Others' Work** – Engineers should not take credit for someone else's work.
10. **Help Fellow Engineers** – They should guide and support others in their profession.

Example:

An engineer refuses a bribe from a company that wants to skip a safety test for their new elevator system.

What is whistleblowing? Explain its types with an example.

Whistleblowing

- **Whistleblowing** is when an employee reports illegal or unethical activities in an organization.
- It helps in preventing corruption and protecting the public.

Types of Whistleblowing:

1. **Internal Whistleblowing** – Reporting the issue within the company.
2. **External Whistleblowing** – Reporting the issue to the media, government, or public.

Example:

An employee at a financial firm exposes fraudulent transactions to protect investors.

7. Note on Professional Responsibilities of Engineers.

Professional Responsibilities of Engineers

1. **Loyalty to the Organization** – Engineers should work in the best interest of their company while following ethics.
2. **Teamwork (Collegiality)** – Engineers should respect and collaborate with colleagues.
3. **Respect for Authority** – Engineers should follow company rules but also challenge unethical decisions.
4. **Commitment to Safety** – Products should be **tested properly** before use.
5. **Honest Communication** – Engineers should **report problems clearly and truthfully**.
6. **Whistleblowing if Needed** – Engineers should report illegal activities in their workplace.

Example:

An aviation engineer finds that a new airplane design has **a serious flaw**. Instead of ignoring it, they report the issue to the authorities to prevent accidents

Here's a one-line summary for each topic:

Right to Professional Conscience – Engineers can make ethical decisions without opposition.

Right to Conscientious Refusal – Engineers can refuse unethical work.

Right to Professional Recognition – Engineers deserve credit for their contributions.

Intellectual Property Rights (IPR) – Protects inventions, brands, and creative works.

Patents – Exclusive rights to an invention for 20 years.

Trademarks – Protects brand names, logos, and symbols indefinitely.

Copyright – Protects creative works like books, music, and software.

Corporate Social Responsibility (CSR) – Companies should help society and the environment.

Safety & Risk in Engineering – Designs must be tested to prevent harm.

Code of Ethics – A set of rules ensuring honesty and fairness in the profession.

IEEE Code of Ethics – Engineers must prioritize safety, honesty, and fairness.

Whistleblowing – Reporting illegal or unethical activities in an organization.

Professional Responsibilities of Engineers – Engineers must be ethical, responsible, and prioritize safety.